

Developing for Falcon Devices

Davide Berti
Software Engineer
Developer Technical Services

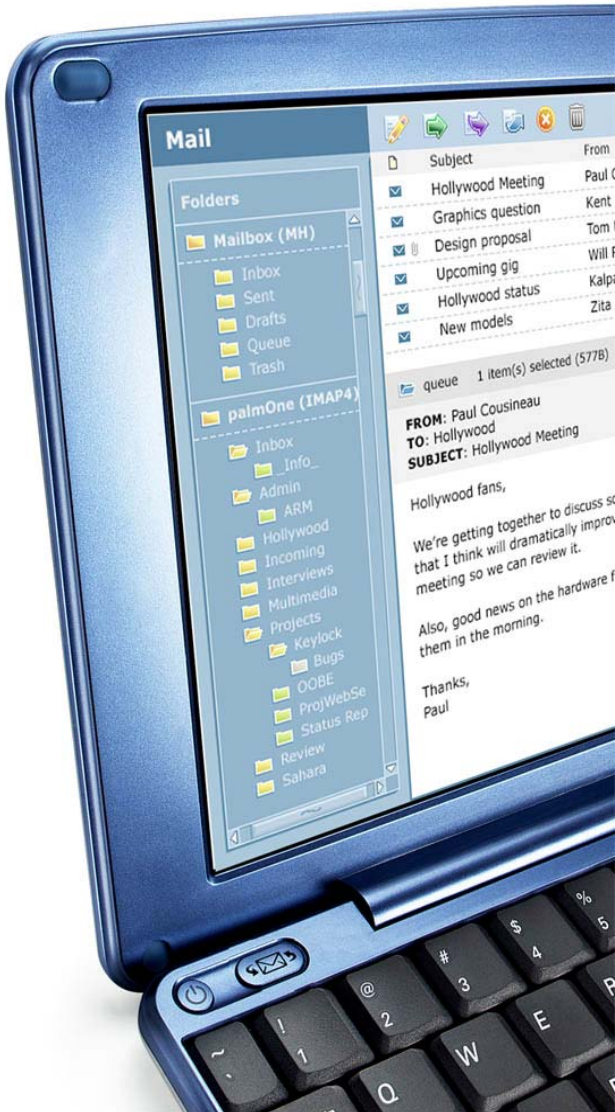
Falcon

A highly mobile companion to a smartphone

- Large color display
- Full size keyboard
- PDA style instant on
- Lightweight
- Email Client
- Documents to Go - Smartphone Sheet, Slide, Word
- Web Browser
- Powerful platform for application developers
- PDF Viewer
- Photo Viewer

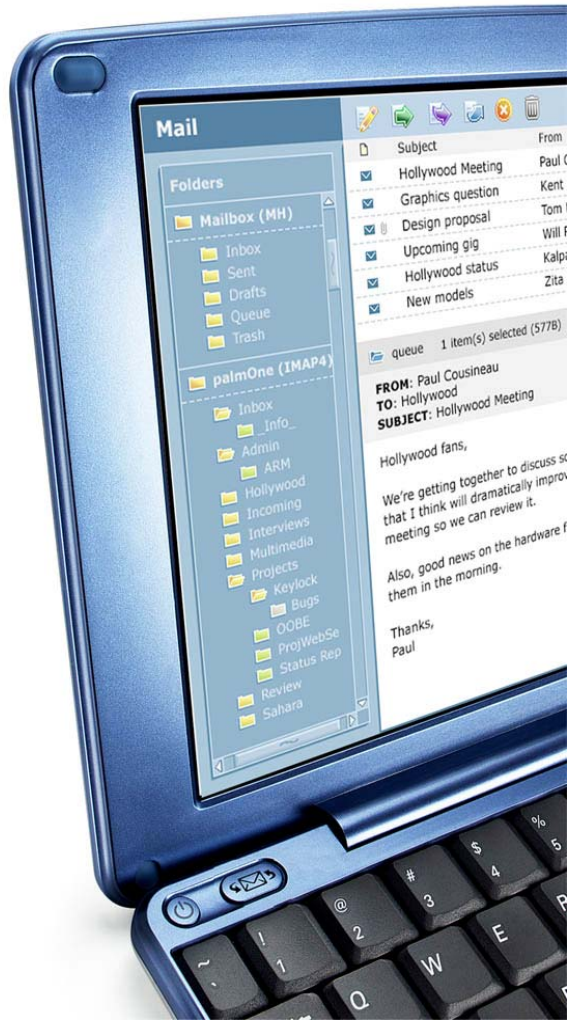
Falcon – Simple. Mobile. Connected.

- Simple
  - ‘Instant On’
  - Seamless wireless email
- Mobile
  - Under 3 lbs, >6 hours battery life
- Connected
  - Able to use anywhere you can make a phone call or where there is a WiFi connection



Palm's Software Focus

- Email application
- Office Documents compatible
  - Word, Excel, PowerPoint, PDF
- Browser

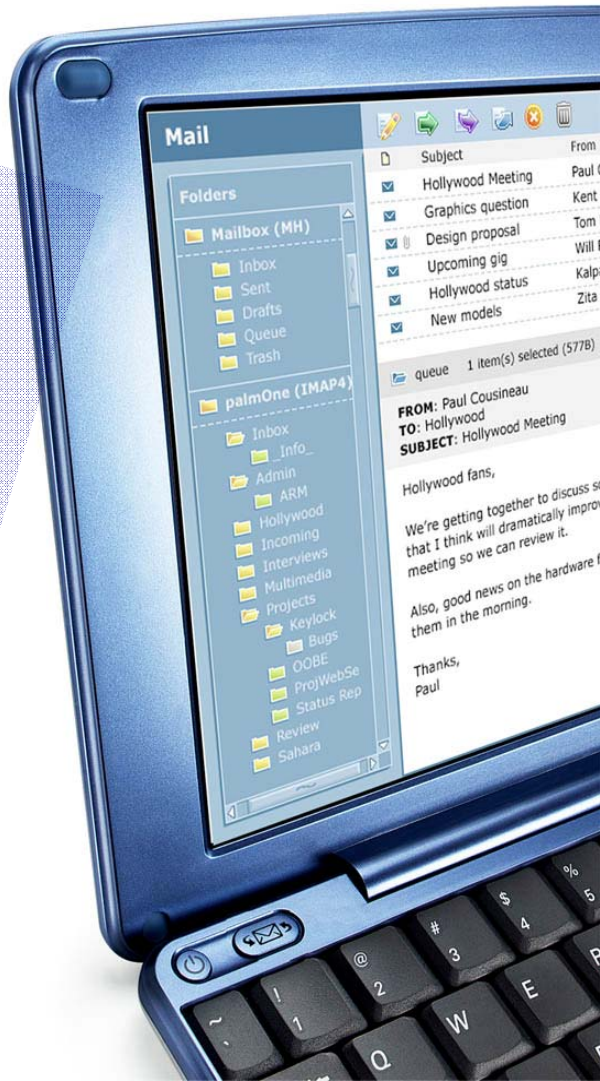


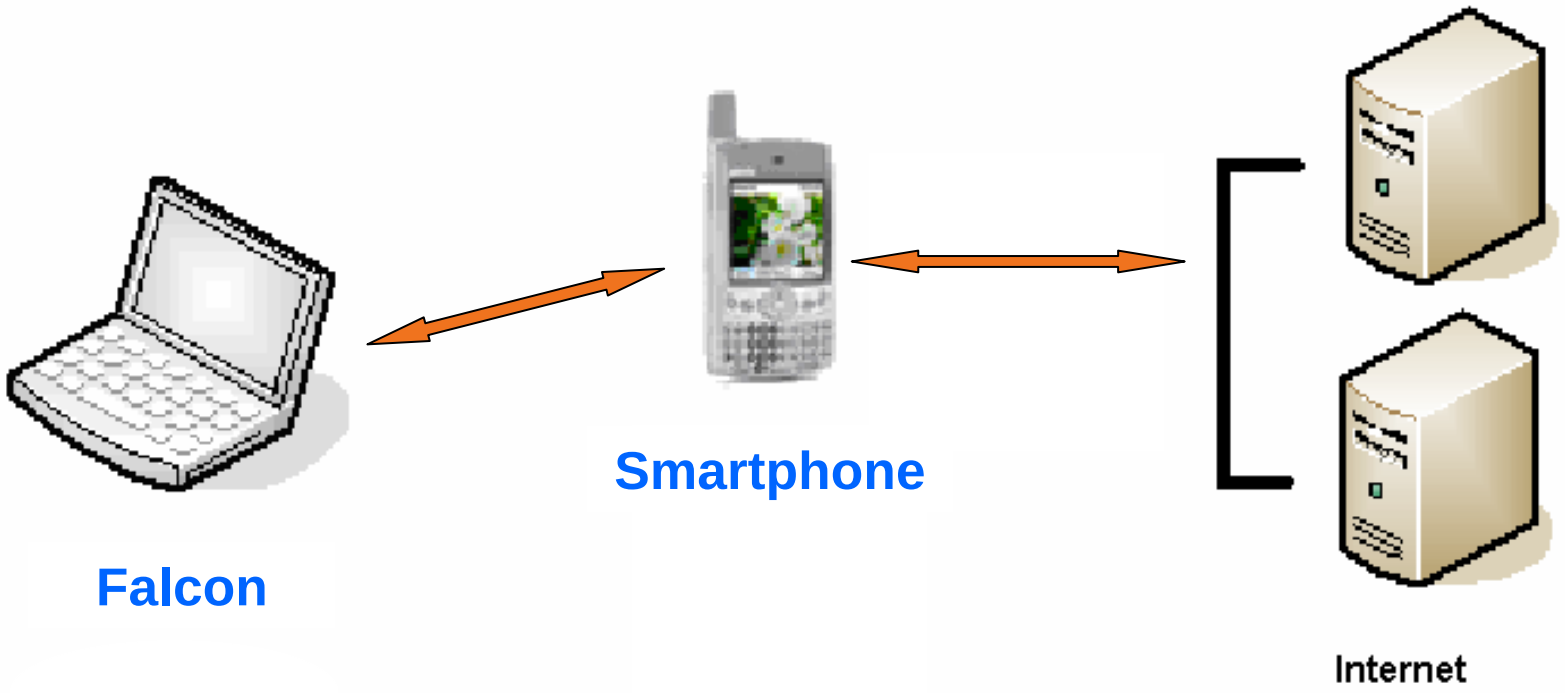
The Smartphone Companion

Falcon is designed to enhance all smartphone devices.





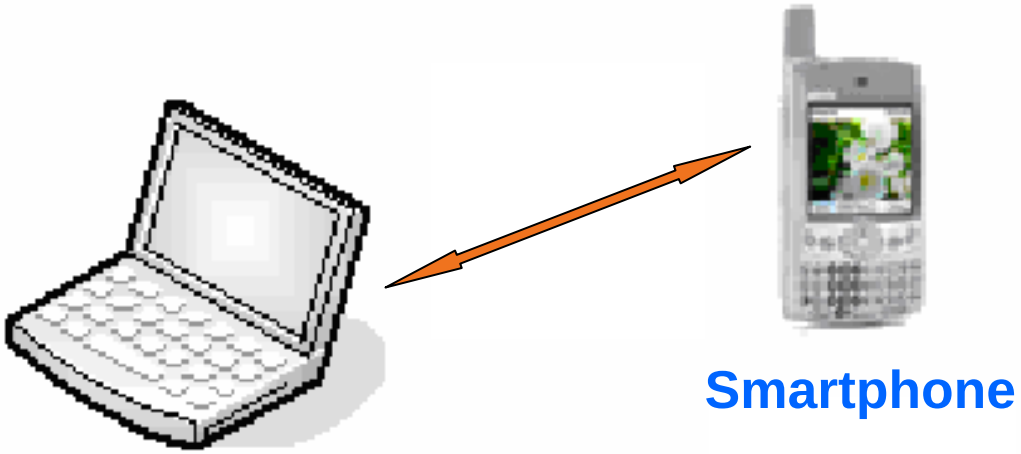




- Enable Bluetooth on the Falcon and the Treo device. Pair the devices*
- Setup a Bluetooth DUN connection to connect to the Internet via the Treo device's data connection.*

* With a device/carrier that supports DUN/Falcon

Falcon Treo Support



Falcon will support a variety of Treo devices.

Treo models 680 and above will be supported

Falcon will also support certain 3rd party WinMo smartphones

Agenda

- Introduction to Falcon
- Falcon Hardware/Software Overview
  - Hardware details
  - System Software
- Getting Started
  - Configuring Ubuntu
  - SDK installation/contents
- Device Specifics/Tools/Utilities
- Debugging
- Software Development
  - Hx System Software
  - HxUI Toolkit
- Appendix



Hardware Features

10.2 inch 1024 x 600 16 bit TFT screen*

18mm pitch keyboard with touchpoint

Track wheel

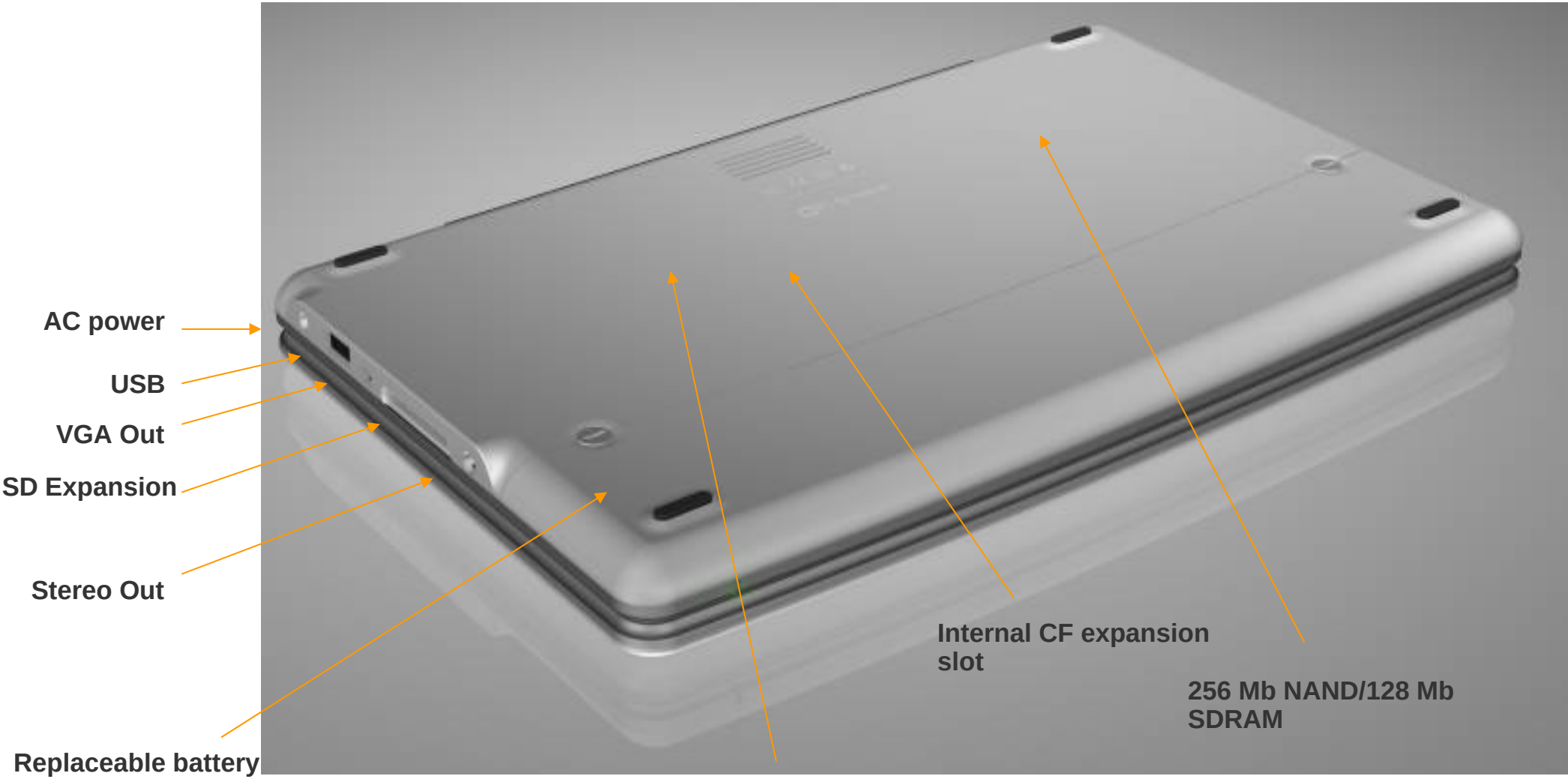
Navigation buttons

Built in WiFi (802.11b) and BT (1.2) radios



*Next Generation Prototype

Hardware Features



Falcon Hardware Overview

Feature	Description
Microprocessor	Intel Bulverde ARM C5e
RAM	64 KB NOR to boot from 128 MB SD RAM to execute in 256 MB NAND store system/apps
LCD Panel	1024 x 600
VGA Video Out	1024 x 768
Battery	8.4 volt Lithium Ion Battery Approx. 4-5 hours of real world use Standby: ~120 days on full charge

Falcon Hardware Overview

Feature	Description
Keyboard	keyboard with Trackpoint cursor control
USB	USB to Ethernet for file transfer, remote debugging
SD Card*	Place to store user documents
Compact Flash	Extra storage
WiFi	Wireless Networks 802.11 b with WEP Personal, WPA-PSK Personal, WEP Dynamic, 802.1x Enterprise Security Standards
Bluetooth	Bluetooth 1.2 Standard Coordinate with Treo, Sync Mail

*Does not support SDIO



Falcon Hardware Overview

Feature	Description
Graphics Chip	Intel® 2700G Multimedia Accelerator Low Power, High Performance Video Acceleration • Accelerated MPEG 2 video decode, •1024 x 600 resolution •Accelerated MPEG 4 video decode •Hardware accelerated Inverse Discrete Cosine •Transforms (IDCT) •Hardware accelerated motion compensation •Hardware accelerated color space conversion •Hardware accelerated scaling Low Power, High Performance Graphics Acceleration •2D and 3D acceleration •Up to 75 MHz, 32-bit graphics engine •831K triangles per second (tps) processing capability, 84M pixels per second (pps) fill rate

Falcon Usage Guidelines

- Alt 5 switches between LCD and video out. Includes VGA Adapter
- Falcon doesn't implement the concept of On/Off.
- Tap power button to turn Falcon off/places it in standby
  - The application will be notified to save its state.
- Tap power button to instantly turn back on.
  - Application is in the foreground
- No reset key; terminate an application.
  - "Apps" + "backspace" will kill the application.
- To reboot the device hold down power button for 5 seconds.
  - Falcon will reboot

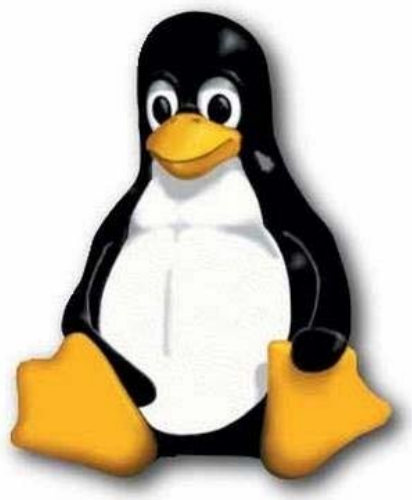


Falcon Usage Guidelines

- Menu emulates the control key
  - Menu ‘c’ is equivalent to the key sequence Control ‘c’
  - Menu ‘[‘ is equivalent to the escape key
- Reboot and hold down the ‘b’ key to get a blob prompt
  - Used to identify the HW revision of the device
  - Update certain hardware specific information if required



Features To Spark Ideas…











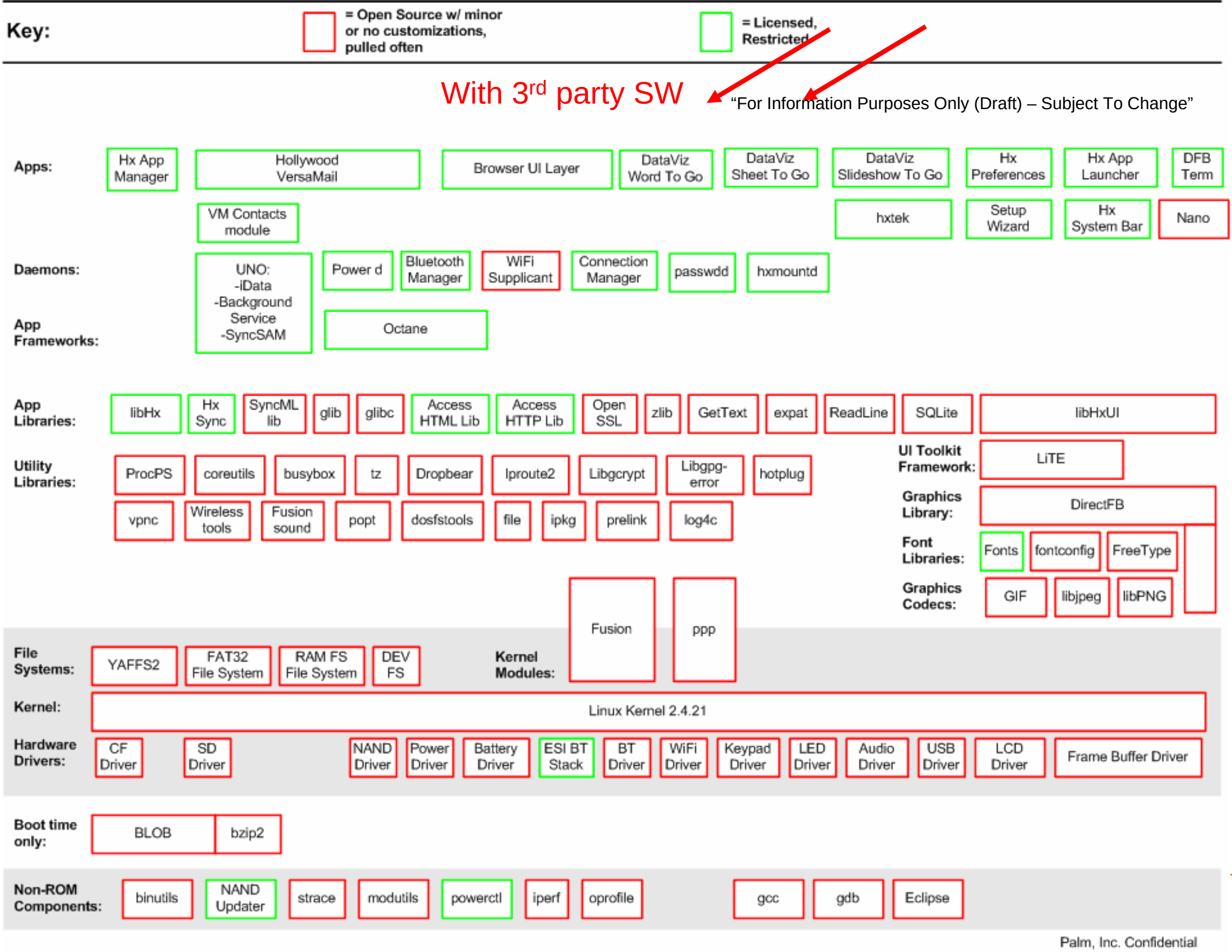




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Palm, Inc. Confidential

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Sans 3rd party SW

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Apps: Hx App Manager | Hollywood VersaMail | DataViz Sheet To Go | DataViz Slideshow To Go | Hx Preferences | Hx App Launcher | DFB Term | hxtek | Hx System Bar | Nano

Daemons: Bluetooth Manager | WiFi Supplicant | Connection Manager | passwdd | hxmountd

App Frameworks:

App Libraries: libHx | SyncML lib | glib | glibc | Access HTML Lib | Open SSL | zlib | GetText | expat | ReadLine | SQLite | libHxUI

Utility Libraries: ProcPS | coreutils | busybox | tz | Dropbear | Iproute2 | Libgcrypt | Libgpg-error | hotplug | vpnc | Wireless tools | Fusion sound | popt | dosfstools | file | ipkg | prelink | log4c

UI Toolkit Framework: LiTE

Graphics Library: DirectFB

Font Libraries: Fonts | fontconfig | FreeType

Graphics Codecs: GIF | libjpeg | libPNG

File Systems: YAFFS2 | FAT32 File System | RAM FS File System | DEV FS

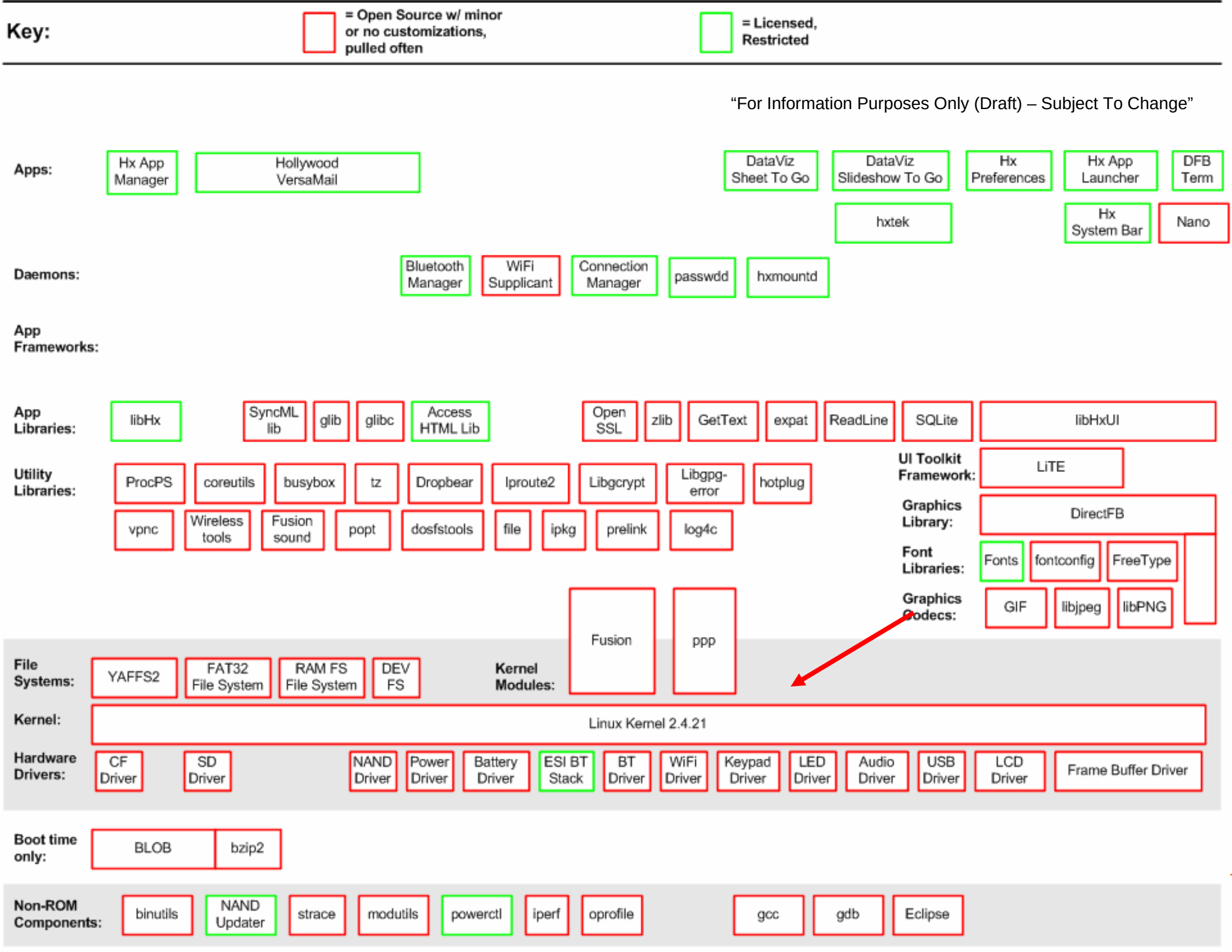
Kernel Modules: Fusion | ppp

Kernel: Linux Kernel 2.4.21

Hardware Drivers: CF Driver | SD Driver | NAND Driver | Power Driver | Battery Driver | ESI BT Stack | BT Driver | WiFi Driver | Keypad Driver | LED Driver | Audio Driver | USB Driver | LCD Driver | Frame Buffer Driver

Boot time only: BLOB | bzip2

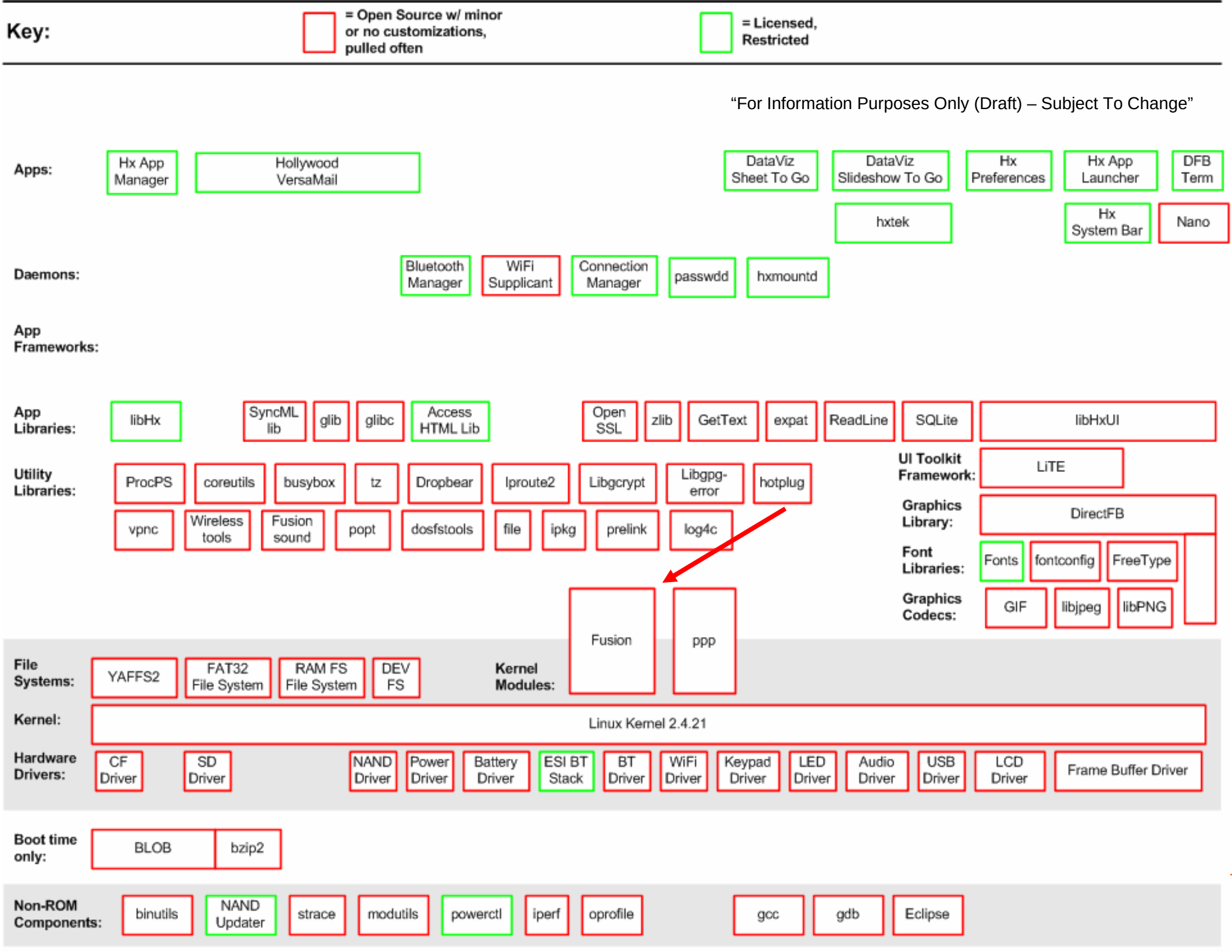
Non-ROM Components: binutils | NAND Updater | strace | modutils | powerctl | iperf | oprofile | gcc | gdb | Eclipse


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Linux Kernel

- 2.4.21-rmk1-pxa1-intc
- power management
- keyboard modification to support custom keys
- SD/MMC/CF changes to support various cards
- changes to support customized memory management
- support for Intel 2700G graphics coprocessor
- support for Fusion IPC
- support for kdb
- support for oprofile
- support for minicore files

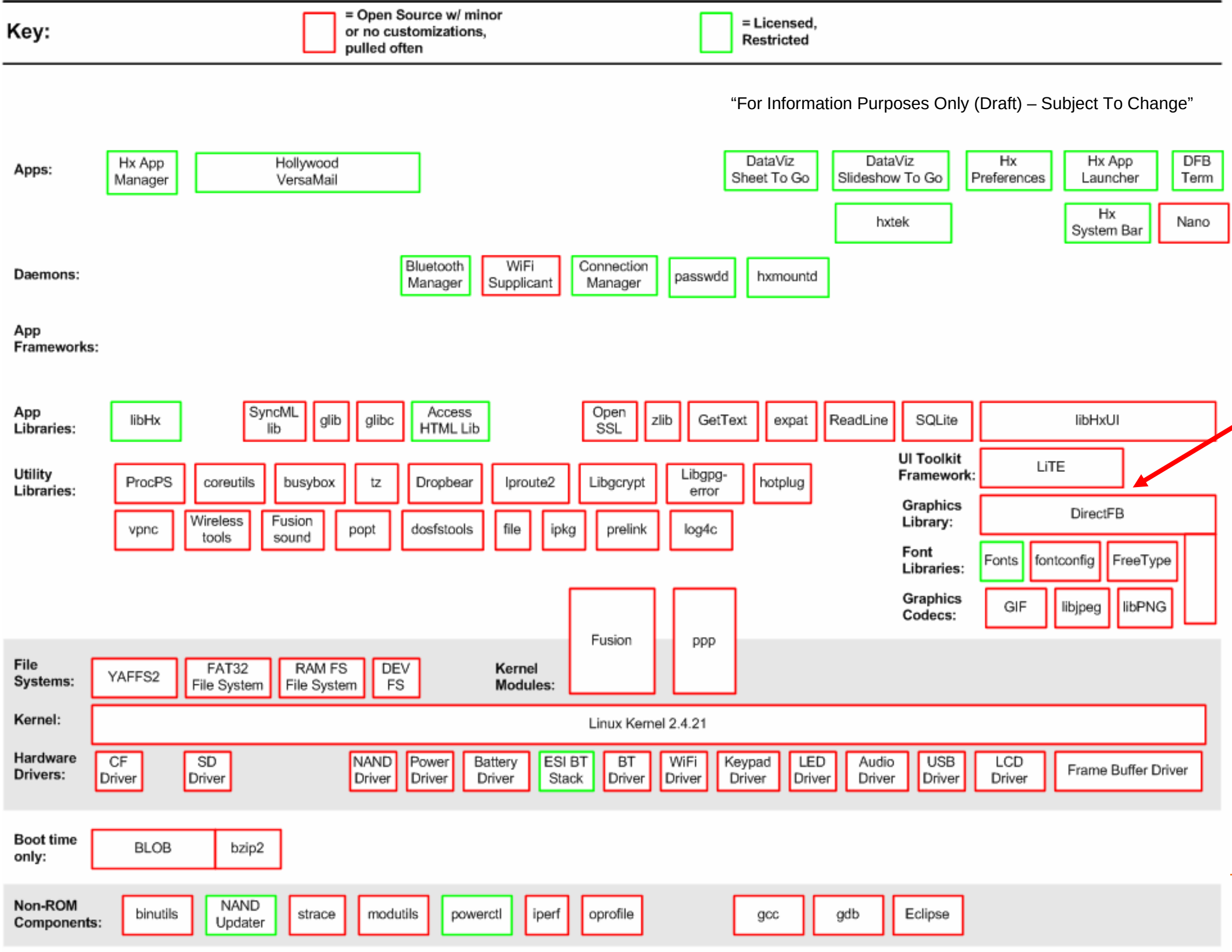



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Fusion

- Fusion is a high level IPC (Inter Process Communication) API providing mechanisms for master/slave environments.
- The master/slave approach employed by fusion eliminates this overhead by virtually merging the processes at the lower level of their implementation. This way of making multiple processes look like multiple threads gave Fusion its name and includes:
- Shared memory provides direct access for all processes to many kinds of shared information, including structures, lists and data. It's also used by the user space part of Fusion itself.




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DirectFB – A New Linux Standard

- DirectFB is a thin library that provides developers with hardware graphics acceleration, input device handling and abstraction, integrated windowing system with support for translucent windows and multiple display layers on top of the Linux Framebuffer Device.
- It is a complete hardware abstraction layer with software fallbacks for every graphics operation that is not supported by the underlying hardware. DirectFB adds graphical power to embedded systems and sets a new standard for graphics under Linux.
- Provides Hardware accelerated graphics for the Intel Marathon Chip





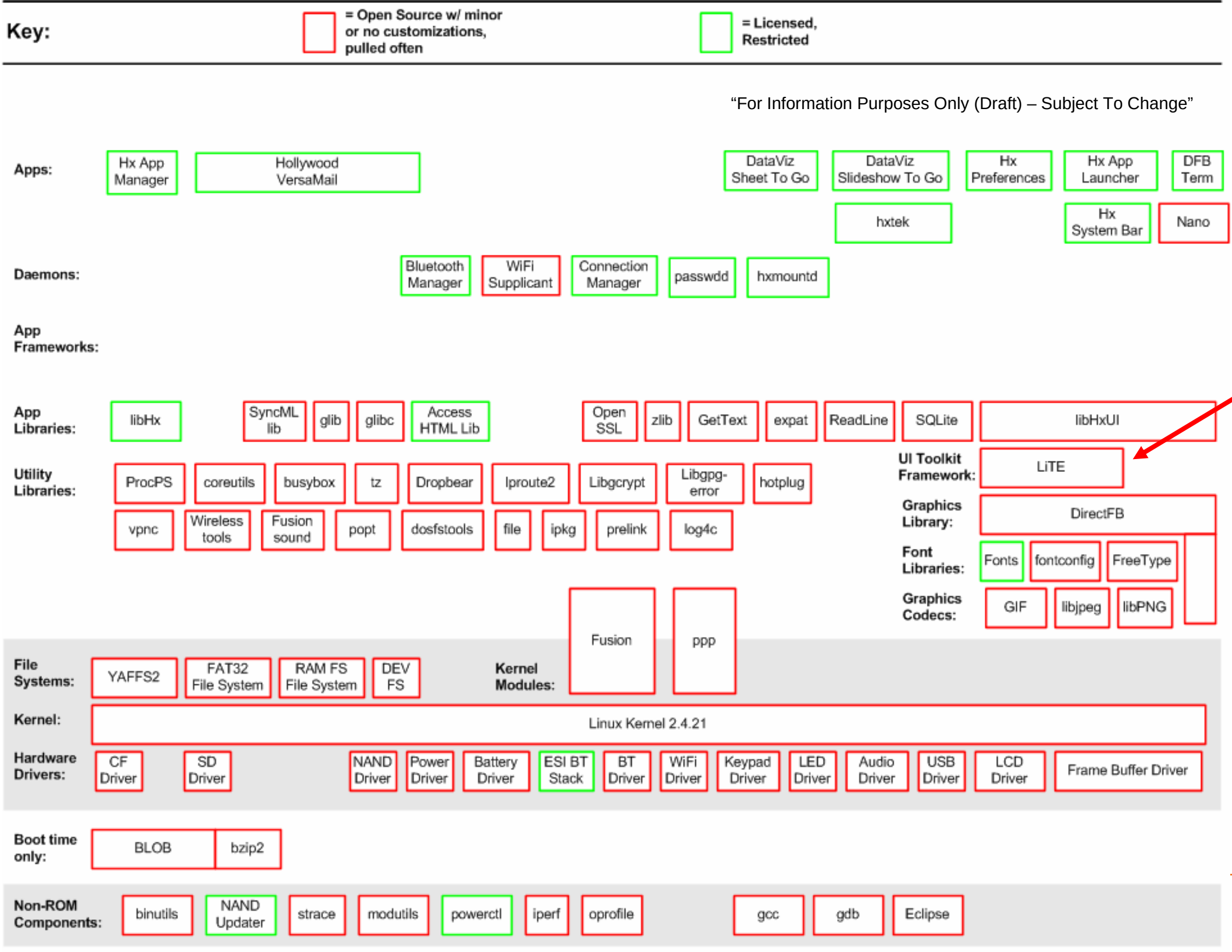
DirectFB – Performance

Excellent Graphics Performance

Fast Blitting

Delivers the Rapid Response Time required by the Platform

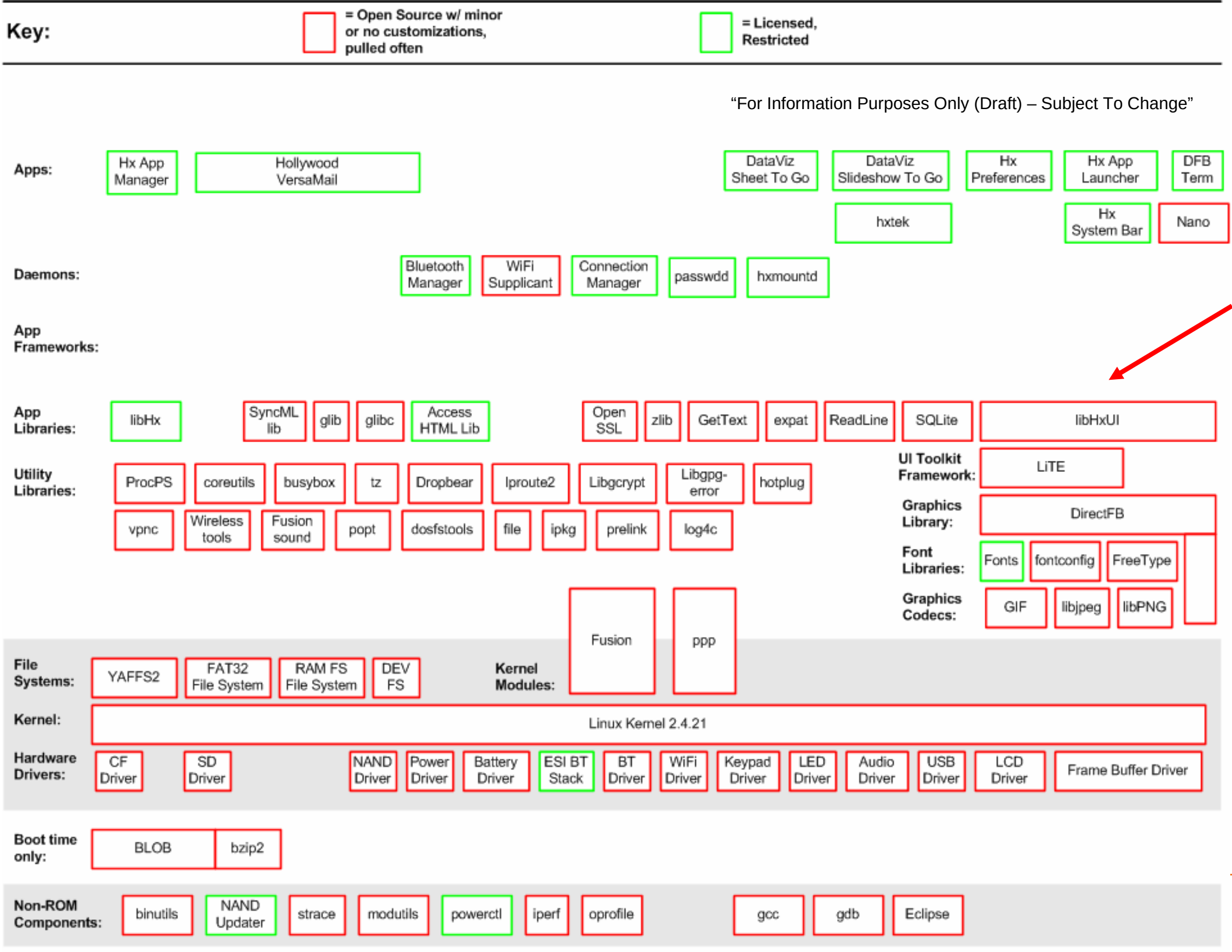



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Lite - Lite is a Toolbox Engine

- Its role is to facilitate the functions of DirectFB so that a toolbox could be written on top of DirectFB with less effort. As such LiTE has abstractions for the underlying graphics and event systems.
- An important abstraction is the LiteBox. It is a low-level view that is tied to a so called sub-surface of a DirectFB surface.
- When a LiTE application initializes itself, as part of that process, a window is most likely created. The window has a DirectFB surface to store the actual pixel content and acts as a top level LiteBox. Any LiteBox created as a child will create a sub-surface.



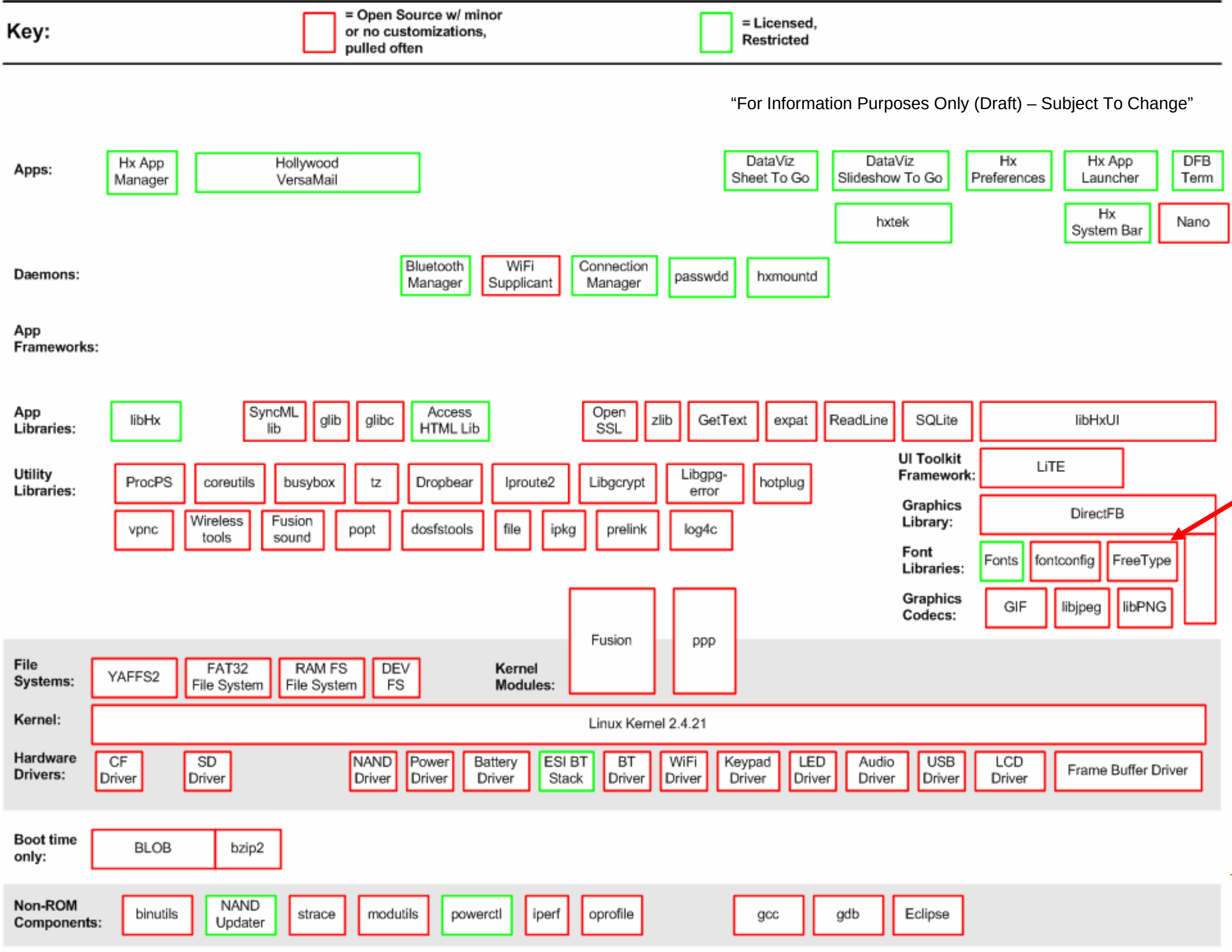

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HxUI – UI Toolkit

The Hxui Toolkit provides the interface/framework for GUI applications.

- Written in C
- Provides an event loop
- XML resource file
- Widget graphical objects
  - Menus, lists, dialogs, etc.
- Provides low-level blitting operations
- Resource ID schema
- Only 1 GUI thread permitted
- Built on top of Lite




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Freetype

- FreeType 2 is a software font engine that is designed to be small, efficient, highly customizable, and portable while capable of producing high-quality output (glyph images). It can be used in graphics libraries, display servers, font conversion tools, text image generation tools, and many other products as well.
- Note that FreeType 2 is a *font service* and doesn't provide APIs to perform higher-level features like text layout or graphics processing (e.g., colored text rendering, ‘hollowing’, etc.). However, it greatly simplifies these tasks by providing a simple, easy to use, and uniform interface to access the content of font files.
- FreeType 2 is released under two open-source licenses: a BSD-like FreeType License and the GPL.



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Boot time only: BLOB; bzip2

Non-ROM Components: binutils; NAND Updater; strace; modutils; powerctl; iperf; oprofile; gcc; gdb; Eclipse

System Fonts

The Default Sunnyvale Font was developed in house. It has been rendered in 2 pixel sizes 13 and 16.

The other System Fonts include:

- Monospace
- Serif
- SansSerif



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Apps: Hx App Manager | Hollywood VersaMail | DataViz Sheet To Go | DataViz Slideshow To Go | Hx Preferences | Hx App Launcher | DFB Term | hxtek | Hx System Bar | Nano

Daemons: Bluetooth Manager | WiFi Supplicant | Connection Manager | passwdd | hxmountd

App Frameworks:

App Libraries: libHx | SyncML lib | glib | glibc | Access HTML Lib | Open SSL | zlib | GetText | expat | ReadLine | SQLite | libHxUI

Utility Libraries: ProcPS | coreutils | busybox | tz | Dropbear | Iproute2 | Libgcrypt | Libgpg-error | hotplug | vpnc | Wireless tools | Fusion sound | popt | dosfstools | file | ipkg | prelink | log4c

UI Toolkit Framework: LiTE

Graphics Library: DirectFB

Font Libraries: Fonts | fontconfig | FreeType

Graphics Codecs: GIF | libjpeg | libPNG

File Systems: YAFFS2 | FAT32 File System | RAM FS File System | DEV FS

Kernel Modules: Fusion | ppp

Kernel: Linux Kernel 2.4.21

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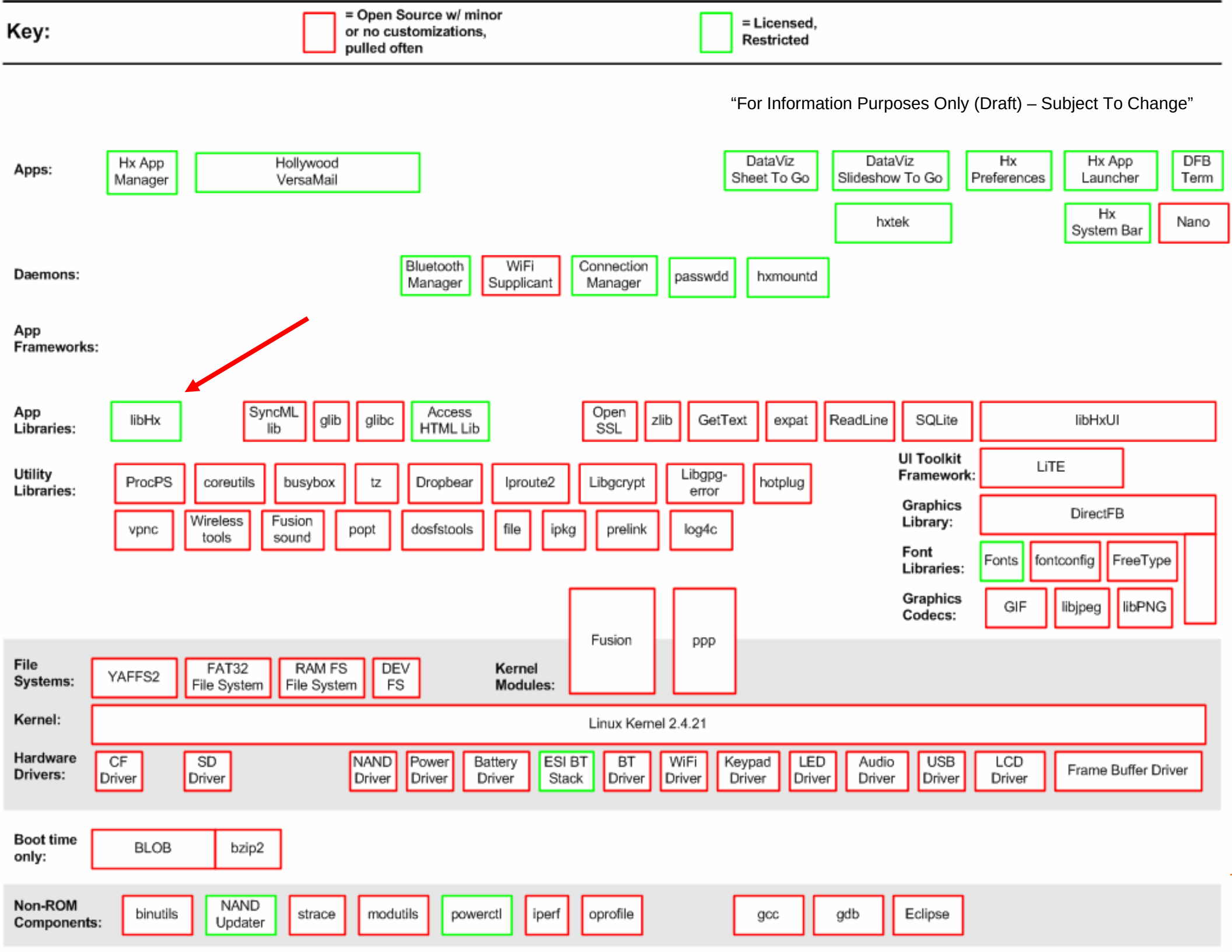
Boot time only: BLOB | bzip2

Non-ROM Components: binutils | NAND Updater | strace | modutils | powerctl | iperf | oprofile | gcc | gdb | Eclipse

Image codecs

- Native support for jpg, png, and gif files is provided by libpng, libjpg, and a gif module
- These are exposed in the HxUI toolkit via a set of image manipulation routines and data structures



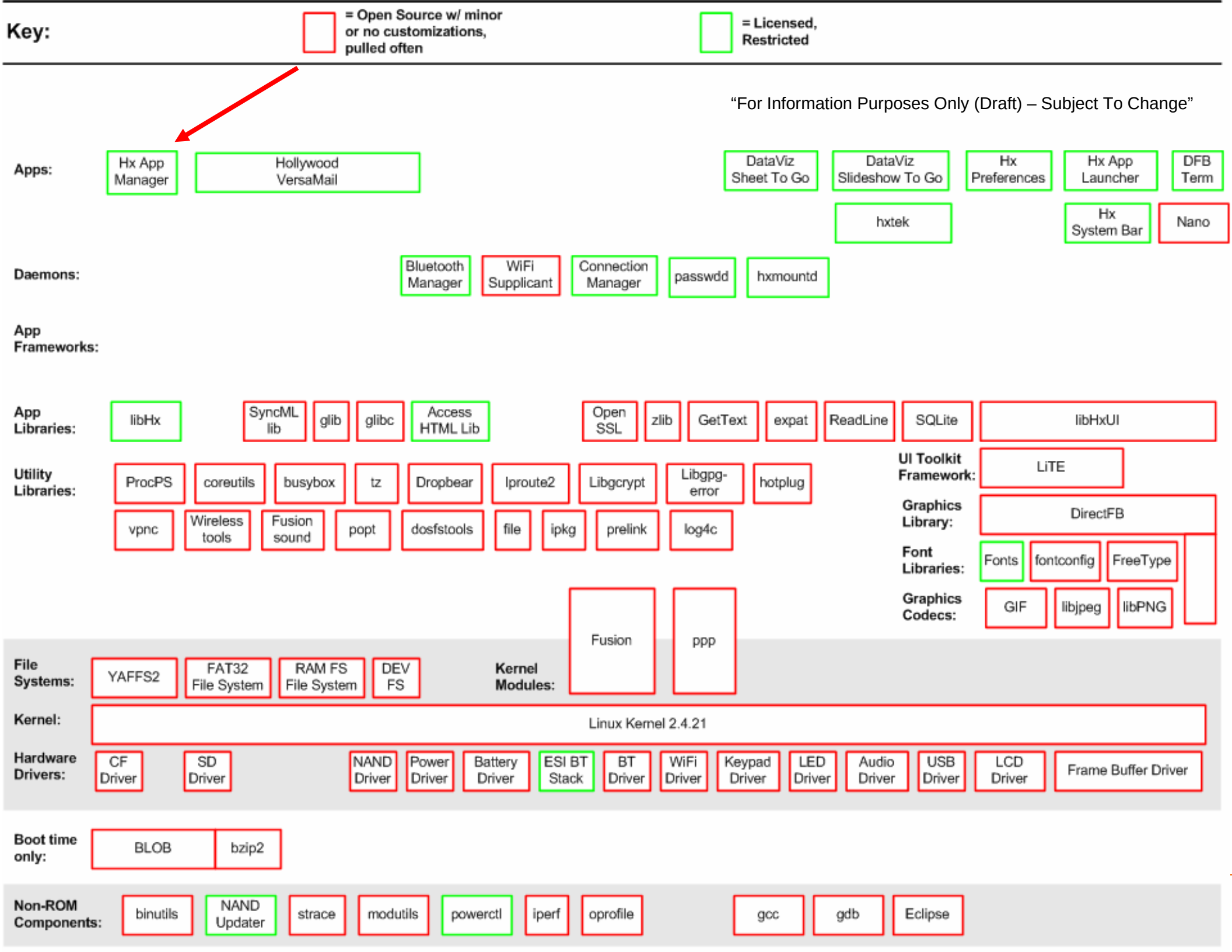

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Hx Library – System Framework

The Hx Library provides a set of resources for applications.

- Written in C
- Provides messaging interface/framework
- Provides system property mechanism
- Power management
- Certificate Manager
- Storage media management

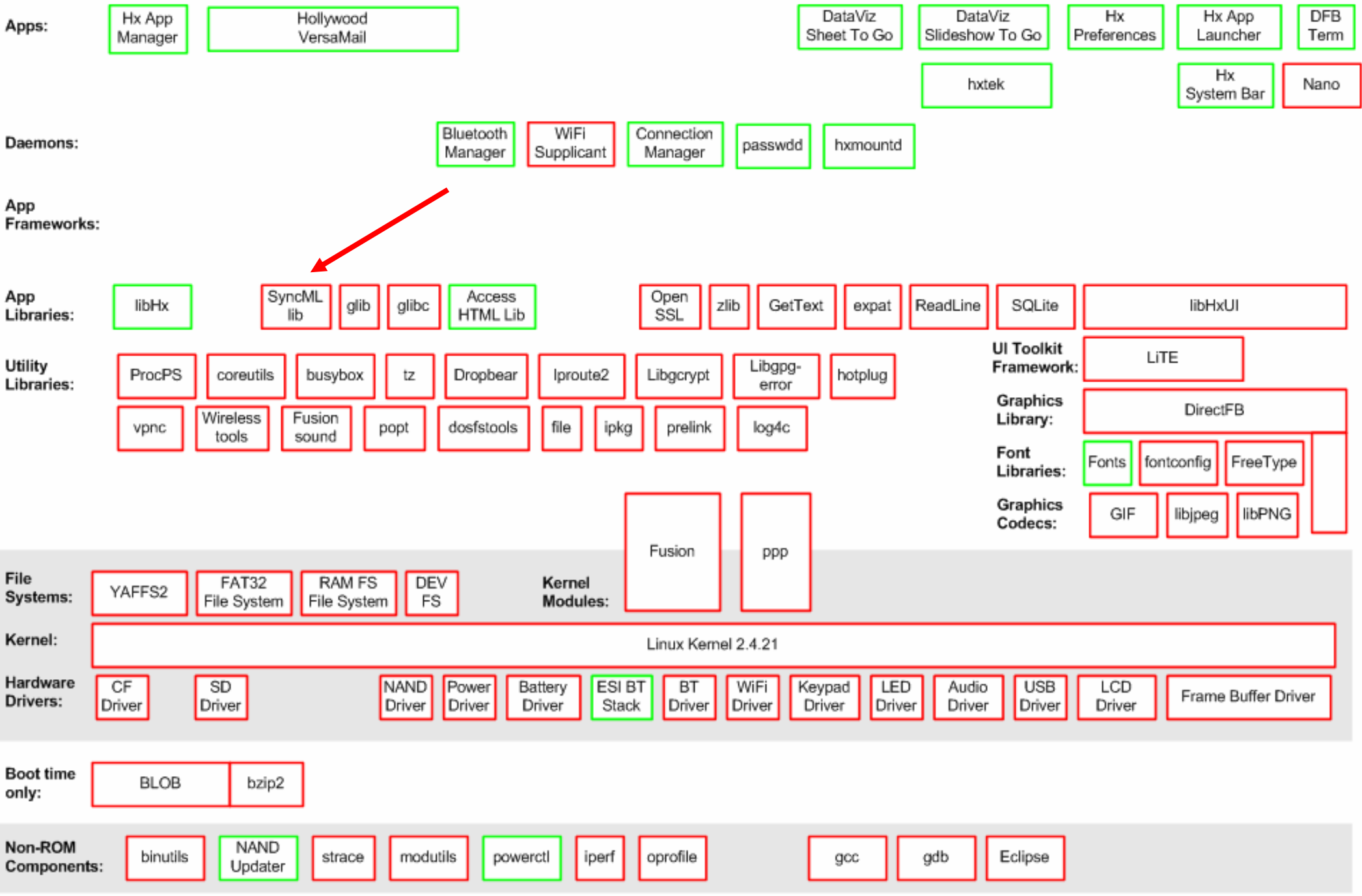



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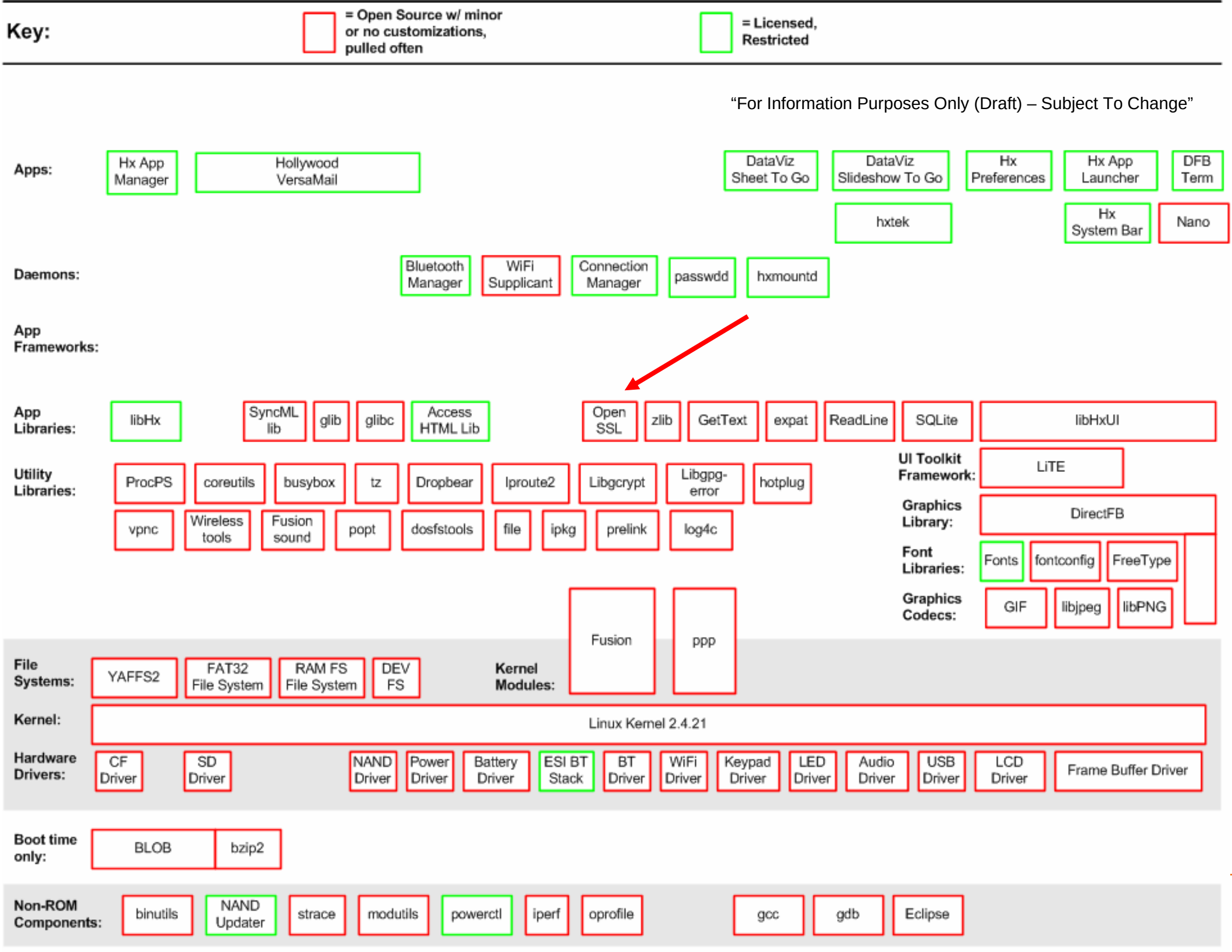
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ACCESS HTML core

This provides an html-rendering core for the HxUI Toolkit.

- Exposed to developers via the HxHtmlView constructs
- Does not provide Flash support
- Can be embedded in an application
- Javascript support




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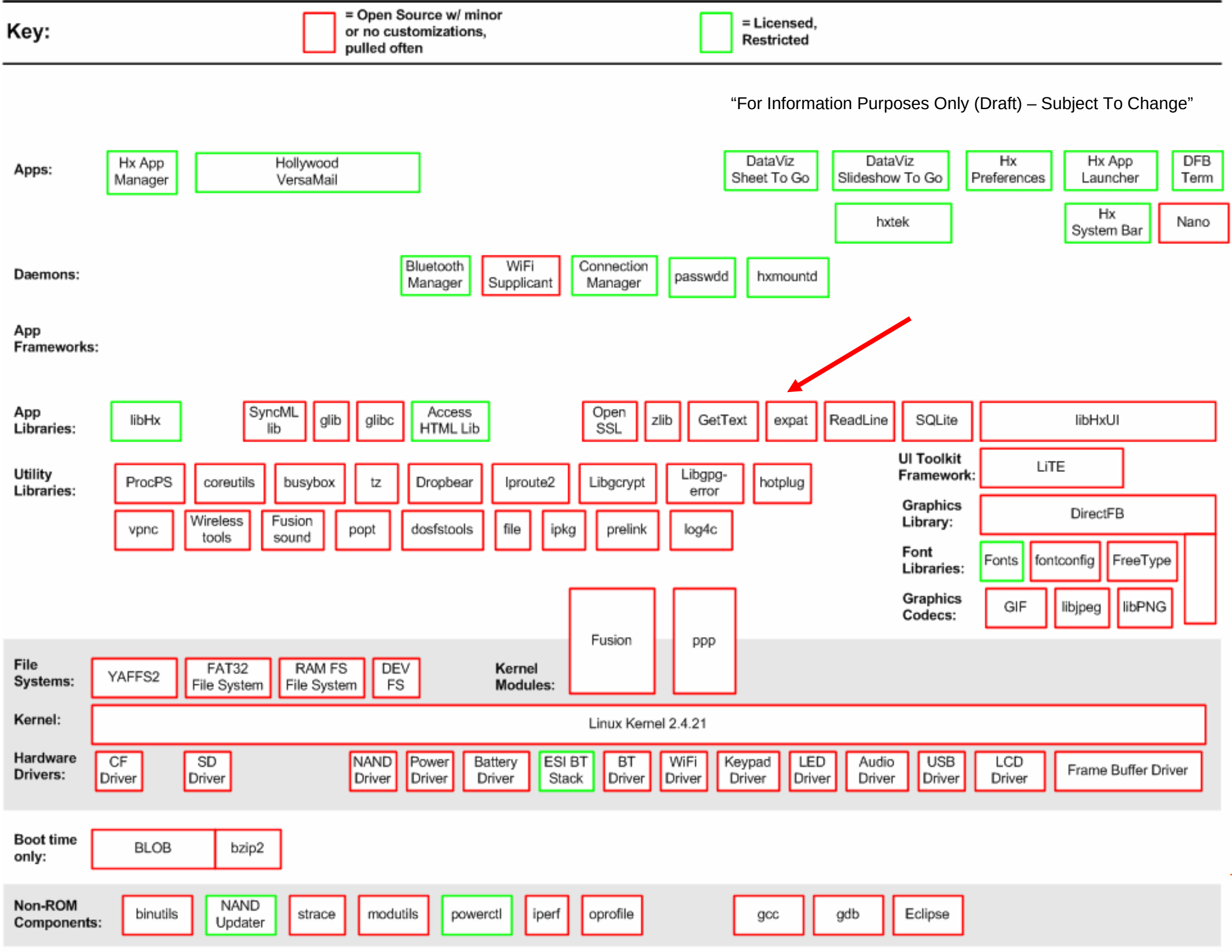
OpenSSL

The OpenSSL Project is a collaborative effort to develop a robust, commercial-grade, full-featured, and Open Source toolkit implementing the Secure Sockets Layer (SSL v2/v3) and Transport Layer Security (TLS v1) protocols as well as a full-strength general purpose cryptography library.

The project is managed by a worldwide community of volunteers that use the Internet to communicate, plan, and develop the OpenSSL toolkit and its related documentation.

OpenSSL is based on the excellent SSLeay library developed by Eric A. Young and Tim J. Hudson. The OpenSSL toolkit is licensed under an Apache-style licence, which basically means that you are free to get and use it for commercial and non-commercial purposes subject to some simple license conditions.



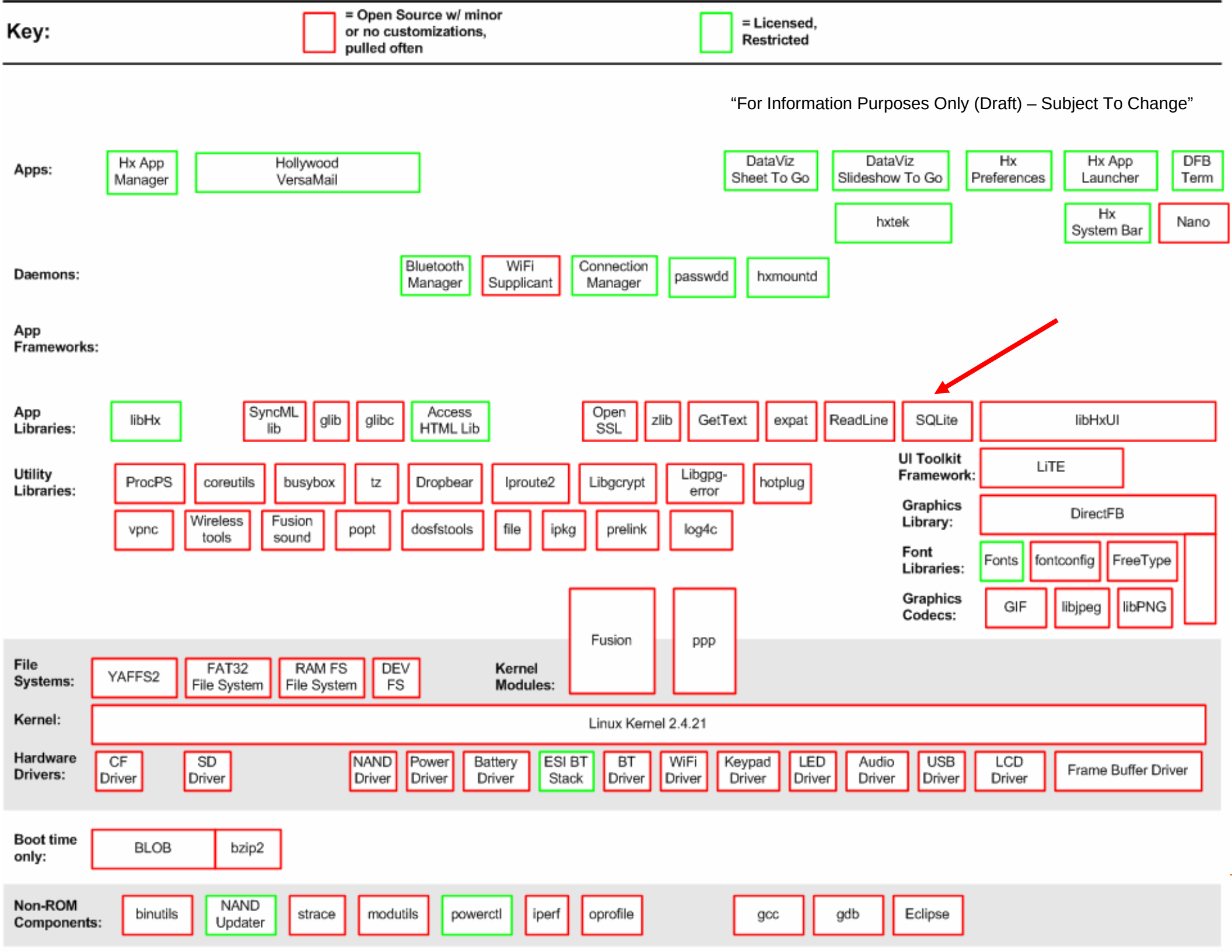

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Expat

Expat is an XML parser library written in C. It is a stream-oriented parser in which an application registers handlers for things the parser might find in the XML document (like start tags).

It's the underlying XML parser for the open source Mozilla project, perl's XML::Parser, and other open-source XML parsers. It’s very fast. It also sets a high standard for reliability, robustness and correctness.




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SQLite

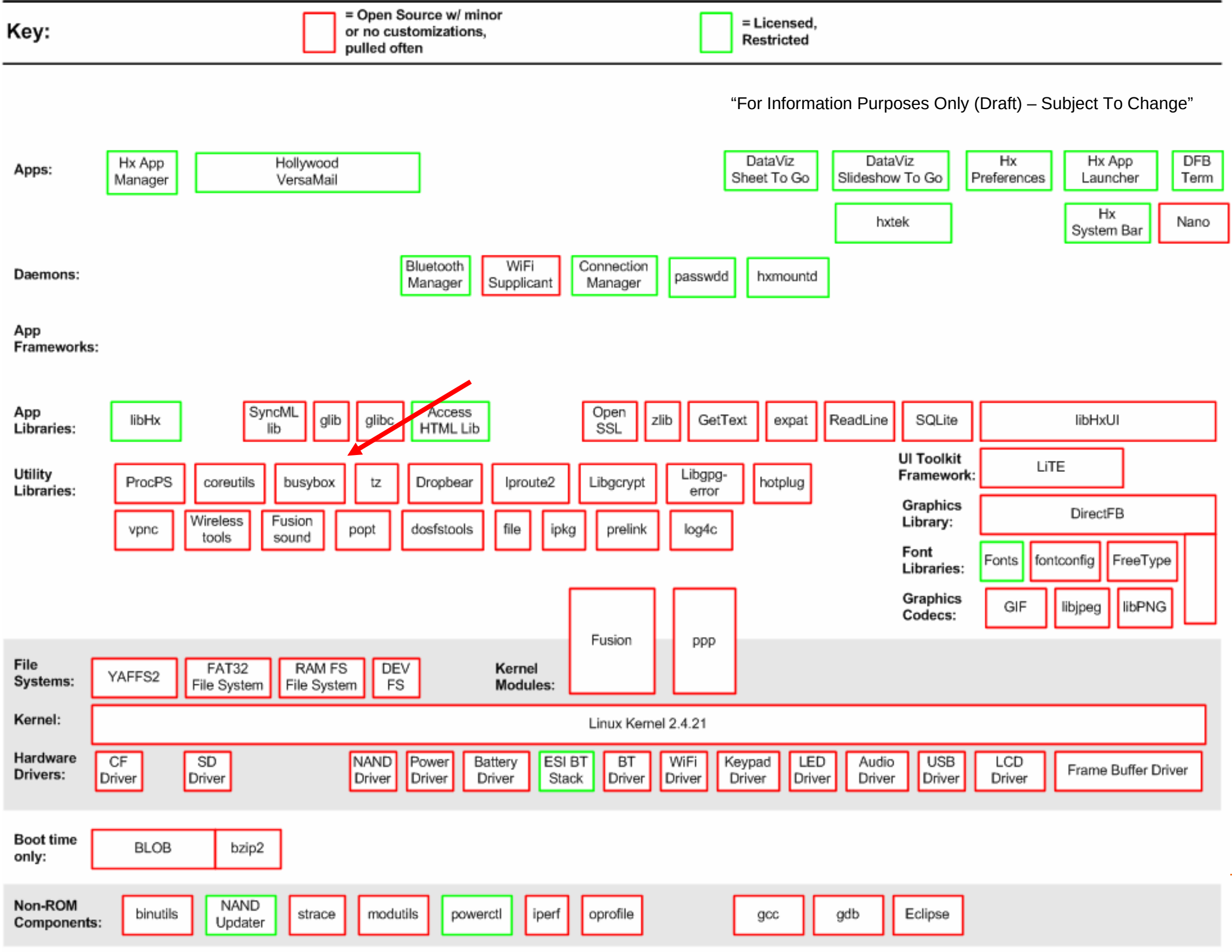
Database Integrity

- Transactions are atomic, consistent, isolated, and durable (ACID) even after system crashes and power failures.

Database Features

- Zero-configuration - no setup or administration needed.
- Implements most of SQL92.
- A complete database is stored in a single disk file.
- Database files can be freely shared between machines with different byte orders.
- Supports databases up to 2 terabytes (241 bytes) in size.
- Sizes of strings and BLOBs limited only by available memory.
- Small code footprint: less than 250KiB fully configured or less than 150KiB with optional features omitted.
- Faster than popular client/server database engines for most common operations.
- Simple, easy to use API.
- TCL bindings included. Bindings for many other languages available separately.
- Well-commented source code with over 95% test coverage.
- Self-contained: no external dependencies. Sources are in the public domain. Use for any purpose.




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BusyBox - Overview

- BusyBox combines tiny versions of many common UNIX utilities into a single small executable. It provides minimalist replacements for most of the utilities you usually find in a traditional unix workstation.
- The utilities in BusyBox often have fewer options than their full-featured cousins; however, the options that are included provide the expected functionality and behave very much like their larger counterparts.
- BusyBox has been written with size-optimization and limited resources in mind, both to produce small binaries and to reduce run-time memory usage.
- BusyBox provides a fairly complete POSIX environment for any small system, both embedded environments and more full featured systems concerned about space. Busybox is slowly working towards implementing the full Single Unix Specification V3

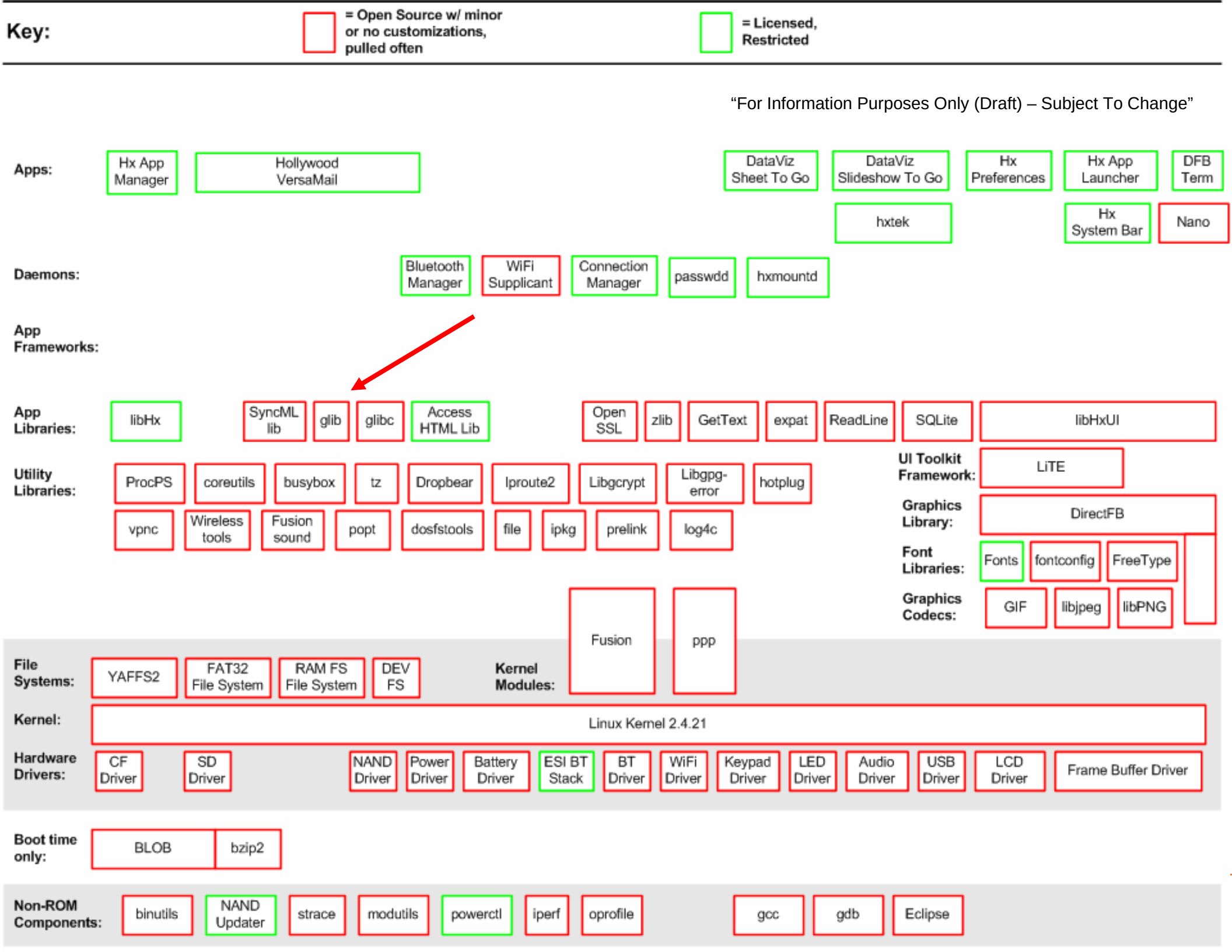


BusyBox - utilities

- bzip2
- coreutils
- File
- Findutils
- Gawk
- Grep
- Inetutils
- Modutils
- net-tools
- Sed
- Tar
- Vim

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Glib

- GLib Utilities
  - String Utility Functions — various string-related functions.
  - Date and Time Functions — calendrical calculations and miscellaneous time stuff.
  - Miscellaneous Utility Functions — a selection of portable utility functions.
  - Lexical Scanner — a general purpose lexical scanner.
  - Automatic String Completion — support for automatic completion using a group of target strings.
  - Timers — functions to time operations.
  - Windows Compatability Functions — functions to support portability to the Windows environment.
- GLib Data Types
  - Memory Chunks — efficient way to allocate groups of equal-sized chunks of memory.
  - Linked Lists — linked lists containing integer values or pointers to data, with the ability to iterate over the list in both directions.
  - Hash Tables — associations between keys and values so that given a key the value can be found quickly.
  - Strings — text buffers which grow automatically as text is added.

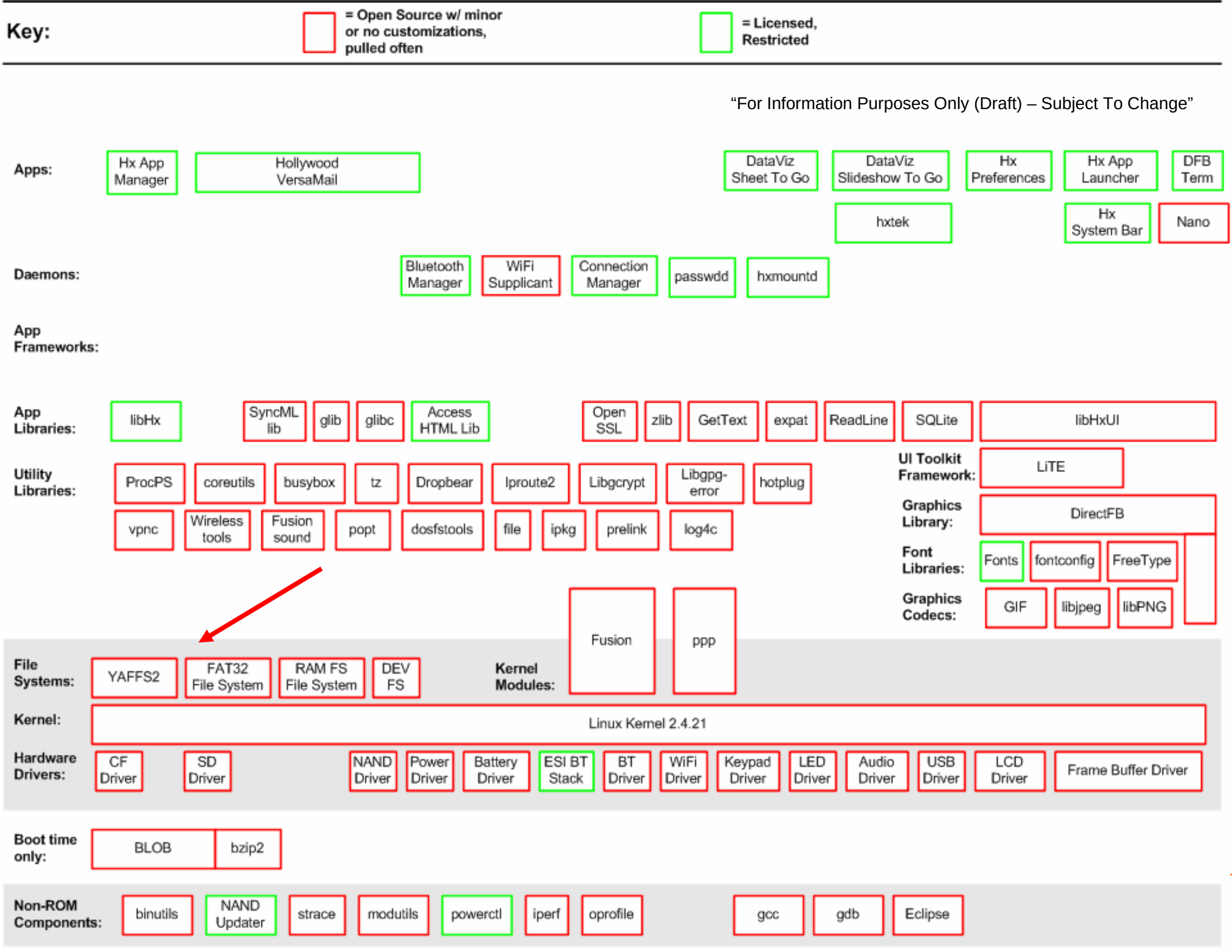
  ...



Glibc

- Any Unix-like operating system needs a C library: the library which defines the ``system calls'' and other basic facilities such as open, malloc, printf, exit...
- The GNU C library is used as *the* C library in the GNU system and most systems with the Linux kernel.
- The GNU C library is primarily designed to be a portable and high performance C library.
- It follows all relevant standards (ISO C 99, POSIX.1c, POSIX.1j, POSIX.1d, Unix98, Single Unix Specification). It is also internationalized and has one of the most complete internationalization interfaces known.




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Falcon System Review

Feature-rich applications can be built with the Falcon Platform

A device with a myriad of possibilities:

- Intuitive UI Toolkit
- Robust Software Packages
- Sophisticated Hardware

